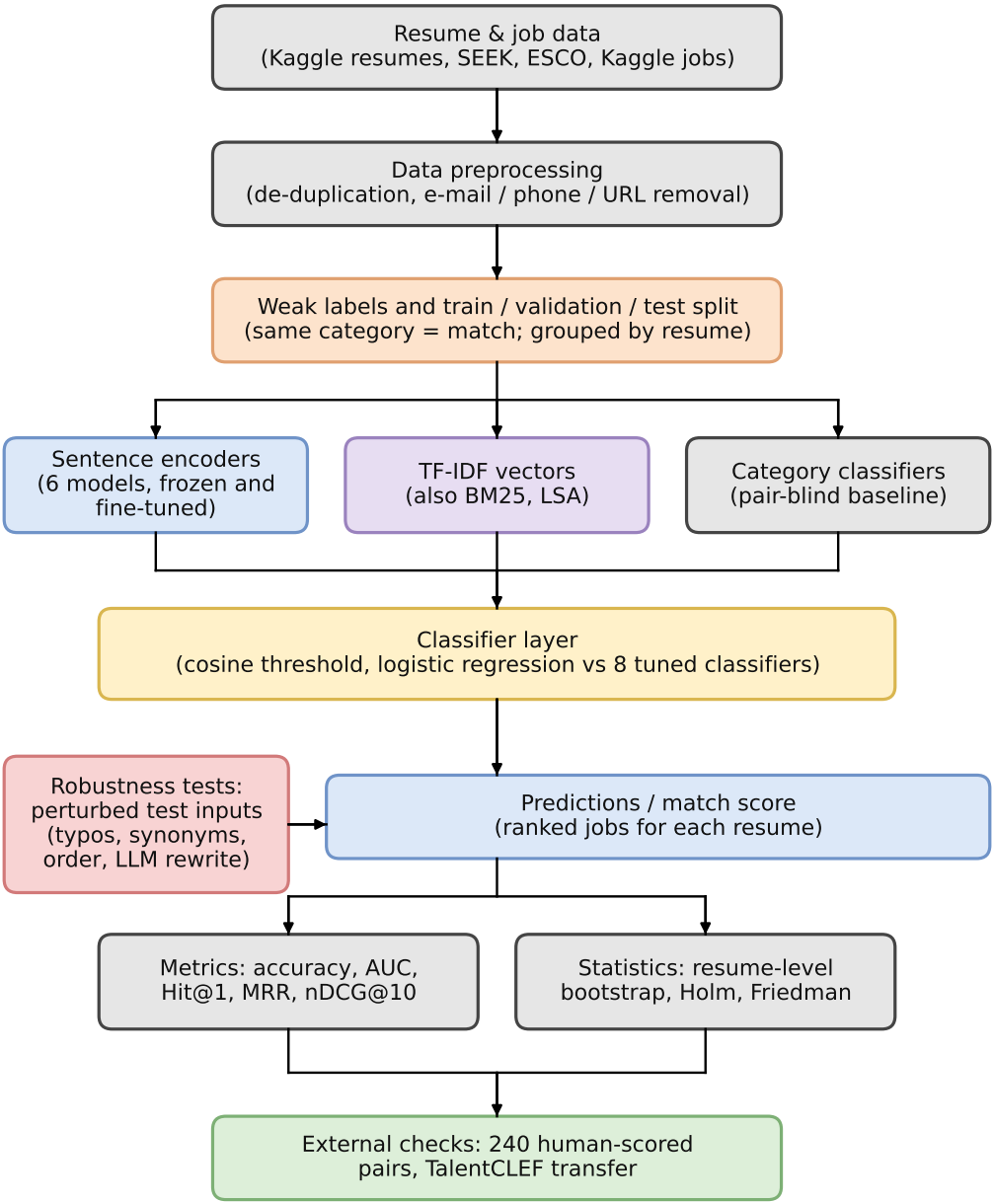
Resume & job data
(Kaggle resumes, SEEK, ESCO, Kaggle jobs)
Data preprocessing
(de-duplication, e-mail / phone / URL removal)
Weak labels and train / validation / test split
(same category = match; grouped by resume)
Sentence encoders
(6 models, frozen and
fine-tuned)
TF-IDF vectors
(also BM25, LSA)
Category classifiers
(pair-blind baseline)
Classifier layer
(cosine threshold, logistic regression vs 8 tuned classifiers)
Robustness tests:
perturbed test inputs
(typos, synonyms,
order, LLM rewrite)
Predictions / match score
(ranked jobs for each resume)
Metrics: accuracy, AUC,
Hit@1, MRR, nDCG@10
Statistics: resume-level
bootstrap, Holm, Friedman
External checks: 240 human-scored
pairs, TalentCLEF transfer